# BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

**MEDICAL DISCLAIMER:** AI-based preliminary estimation only. This result must be confirmed using a standard laboratory blood typing test before medical use. This system does not replace standard serological blood typing.

## 1. EXAMINATION & SAMPLE AUDIT DETAILS

| | | | |
|---|---|---|---|
| **Report ID:** | BG-000301 | **Timestamp:** | 2026-09-06 19:06:06 |
| **Input Source:** | Upload | **Evaluation Status:** | **LOW_CONFIDENCE** |
| **Model Name:** | ResNet50 Transfer Learning | **Confidence:** | **53.60%** |

## 2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP



# Predicted Blood Group:

# O-

Confidence Score: **53.60%**

Threshold: 60.0%

Classification: Low Confidence - Retest Required

Derived ABO Group: **O** (89.8% prob) | Rh Factor: **-**

**Rh FACTOR UNCERTAINTY ADVISORY:** Rh factor (±) determination is uncertain (narrow Rh probability margin (17.4%); moderate 8-class confidence (53.6%); predicted Rh-negative requires serological validation due to weak ridge-Rh correlation). ABO group appears to be 'O' (89.8% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

## 3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

| A+ | A- | AB+ | AB- | B+ | B- | O+ | O- |
|---|---|---|---|---|---|---|---|
| 1.7% | 3.9% | 0.8% | 3.8% | 0.1% | 0.0% | 36.2% | 53.6% |